

Theory of Computing

Midterm Solutions

2023/2024

2ND YEAR - ENSIA

MARKING NOTES

- First exercise, we have been very+ flexible with the answers provided by the students. Turning a blind eye on the missing transitions to the Trap state....
- Finding the word "accept" means it is not fully correct, but it was accepted.
- For most questions, the answers do not need to match the provided/suggested solutions, we find a counterexample to the solution provided by the student.
- A max bonus of 1.5 point is given to students if there are at least 3 correct answers (0.5 point per question)

EXERCISES : Answer all questions.

Exercise 1 (8 points) [DFA & NFA] :

1. [1 point] Describe the language for the following machine :

All words for the exception of a and b

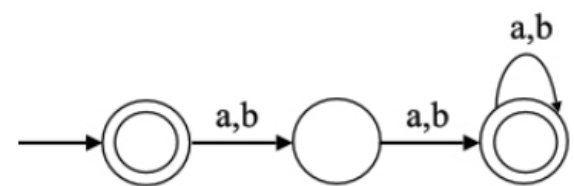
All words except $|w|=1$

All words except words of length = 1

$|w| \geq 2 + \text{epsilon}$

Epsilon $\{a,b\}^*$

Or any other sentences that seem right...

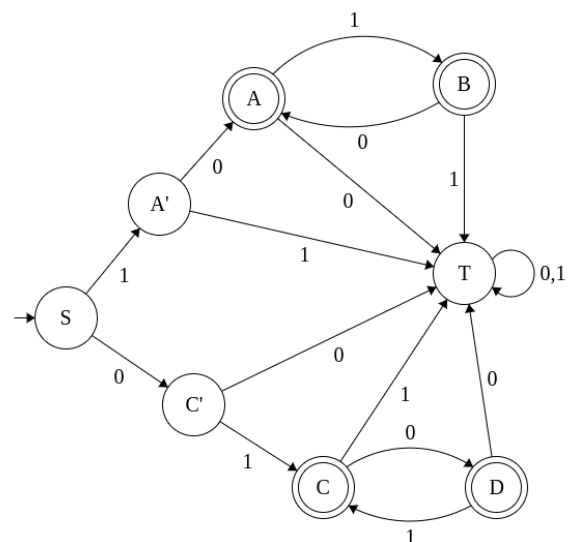


2. [2 points] Draw the DFAs for the following language over the alphabet $\{0,1\}$:

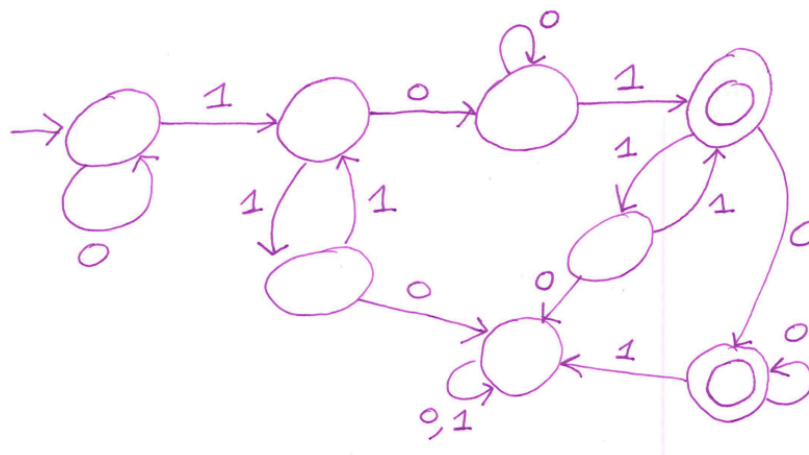
a. Alternating 0 and 1

01

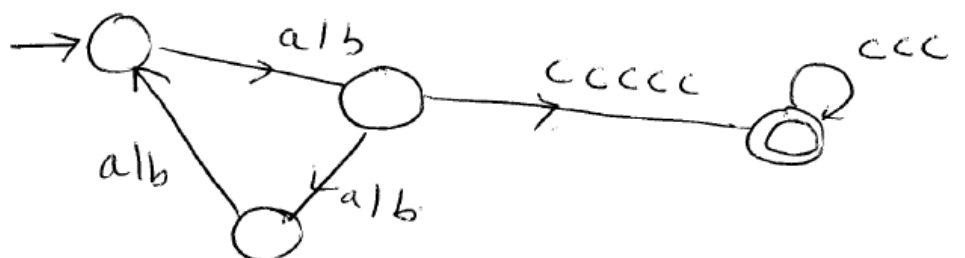
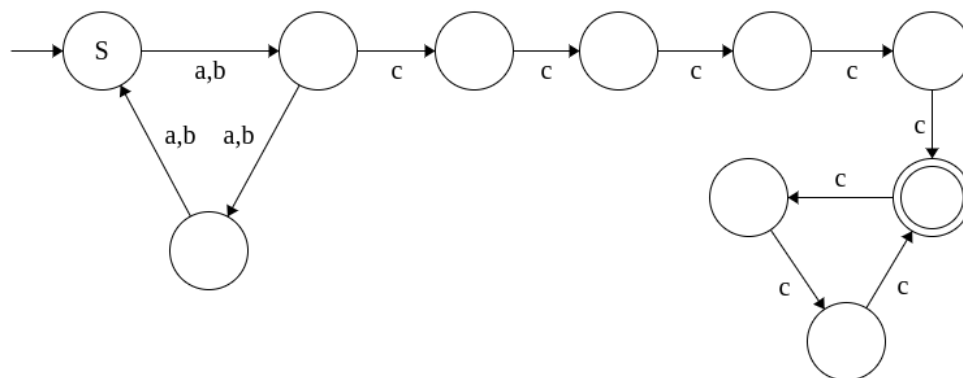
10101



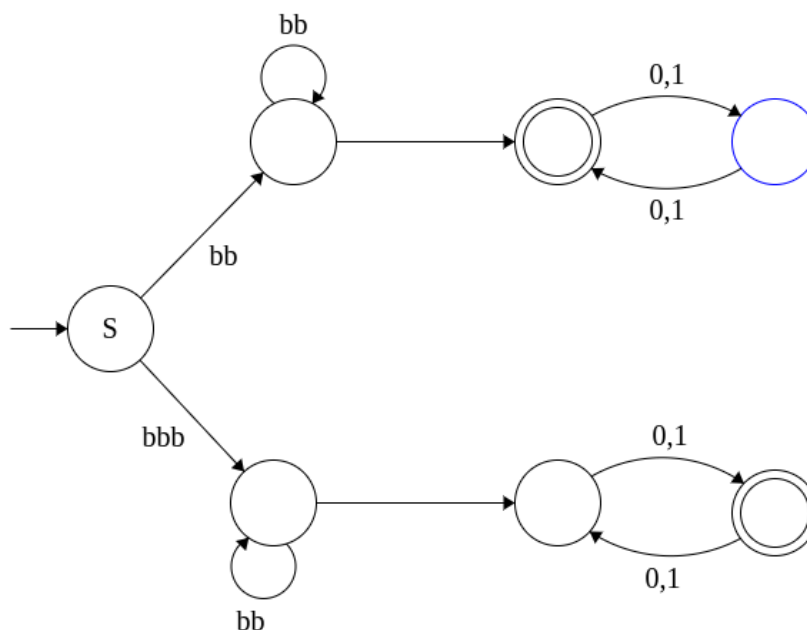
- b. The language where the word contains exactly two blocks of 1's and both these blocks have odd length. For example, 1010 and 01001110000 should be accepted, but not 00110001 or 10101.



a. $\{wc^{3n+2} \mid w \in \{a,b\}^*, |w| \bmod 3 = 1 \text{ and } n \geq 1\}$ (Length of w modulus 3 is 1),
The alphabet is $\{a,b,c\}$

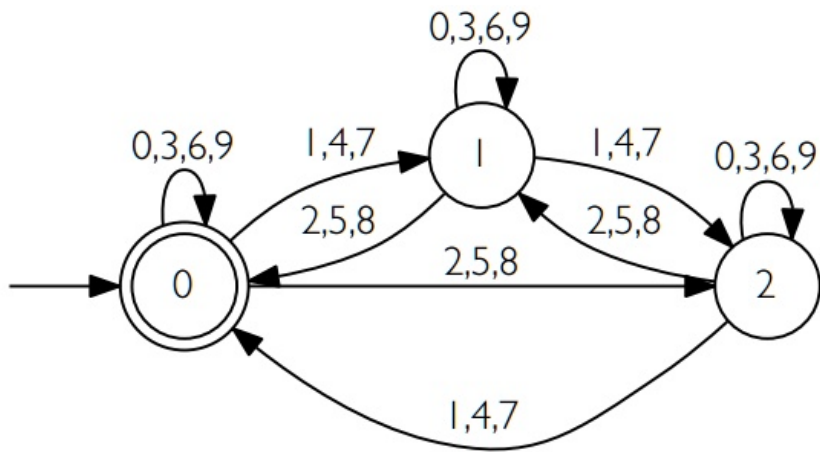


Can be simplified by considering Even $|B^*|$ + Even $|w|$ **union** Odd $|B^*|$ + odd $|w|$

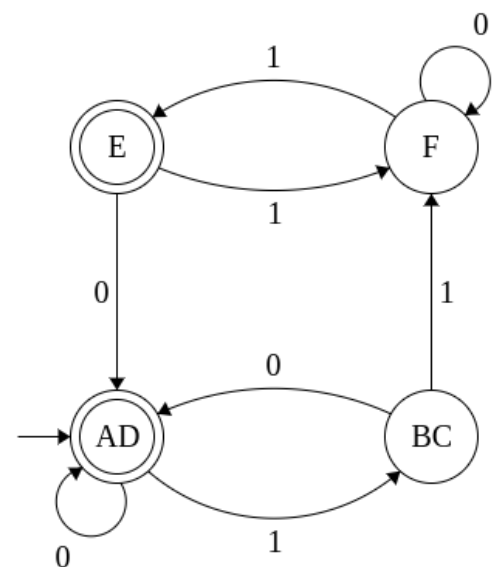
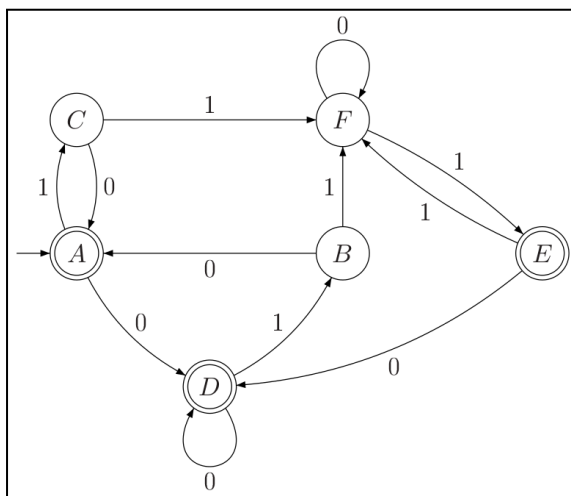


4. [2 points] For the alphabet $\{0,9\}$, construct the DFA for the language where the sum of the digits is a multiple of 3.
(For the solution below, the Epsilon is accepted too which shouldn't be, for simplicity, we assume it can be considered.)

Students who draw the transitions table, is an accepted solution too.



5. [1 point + 1 Point Bonus] Minimize the following DFA, no need to show steps :



Exercise 2 (4 points) [Regular Expressions] :

1. [3 points] Write regular expressions for the following regular languages, the alphabet is $\{0,1\}$

a. [1 point] All words which start and end with the same letter
 $0(0|1)^*0 \mid 1(0|1)^*1 \mid 0 \mid 1$ (even though the ϵ is philosophical , as the start is the same as the end)

We accepted even the solution : $0(0|1)^*0 \mid 1(0|1)^*1$ (which is incorrect)

b. [1 point] Words must start and end with '1'. Any '0' in the word must be immediately preceded and followed by '1'. The length of the word must be at least 4.

$(1111 \mid 10101 \mid 1101 \mid 1011) (01|1)^*$

$1 (1|01)(1|01)(1|01)(1|01)^*$

$(111|101) (1|01)^+ \mid 1101(1|01)^*$

Simulated Strings : 1111 ,1101, 1011, 11101, 10101, 10101...

Strictly Marked

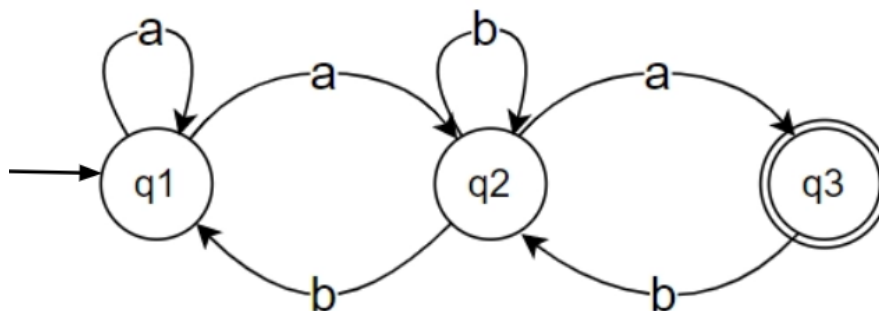
c. [1 point] The language of all strings not containing the substring 000.
 $(1 \mid 01 \mid 001)^* (\epsilon \mid 0 \mid 00)$

or

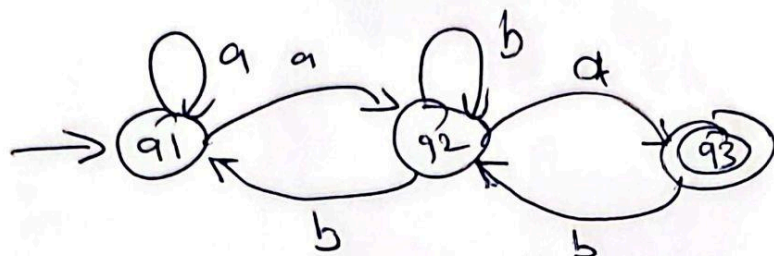
$1^* (011^* \mid 0011^*)^* (\epsilon \mid 0 \mid 00)$

Strictly Marked

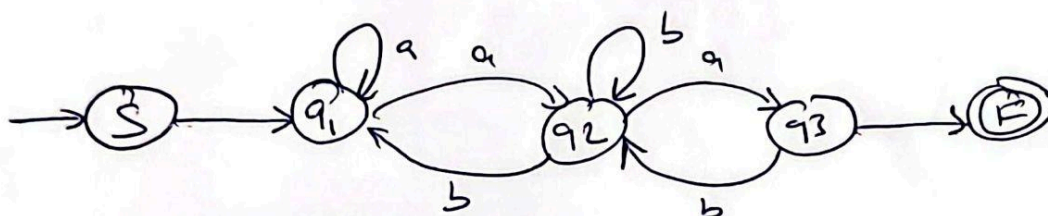
2. [1 points] Convert the following NFA to regular expressions showing the steps



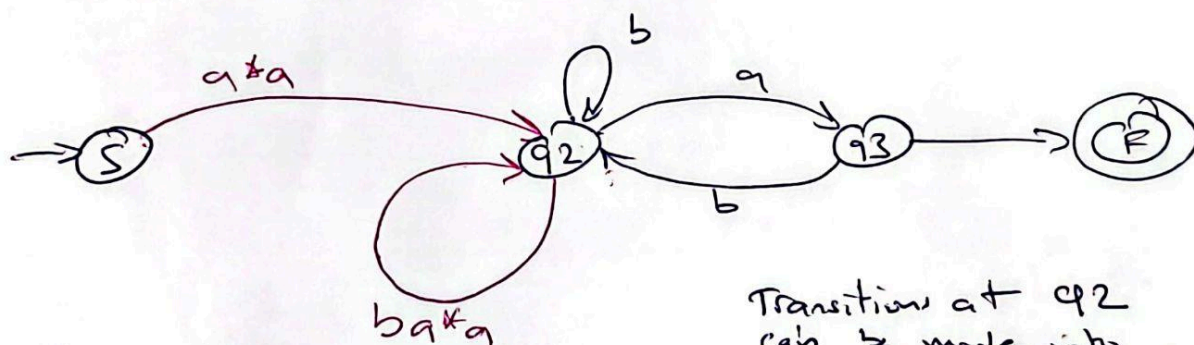
Non-optimized solutions are not accepted (due to doubling the self transitions...)



①

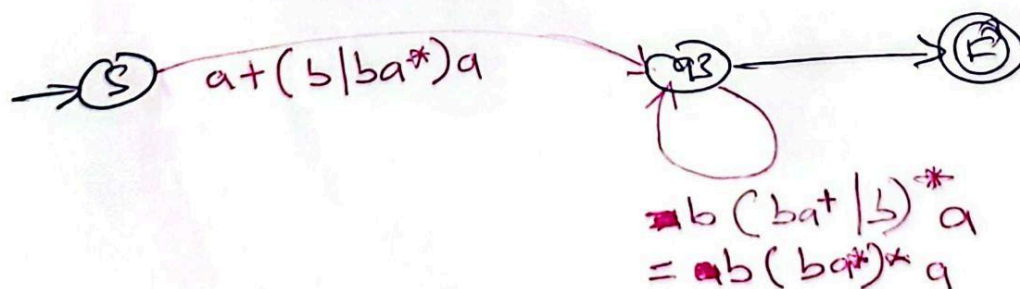


remove q_1

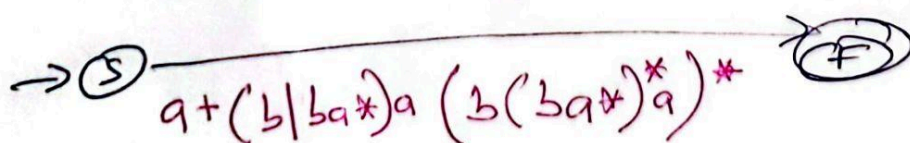


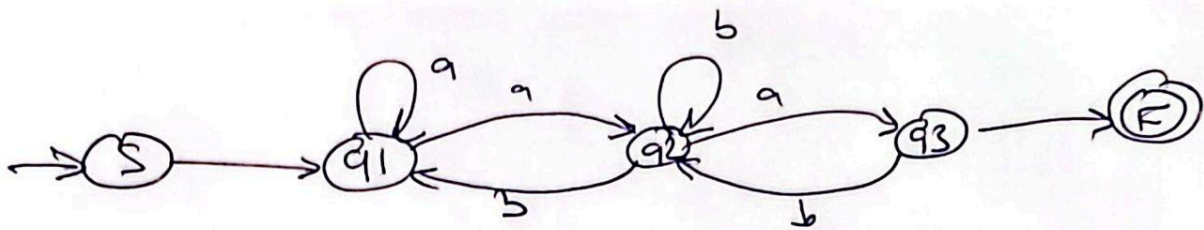
Transition at q_2
can be made into
one transition by | (or)

remove q_2

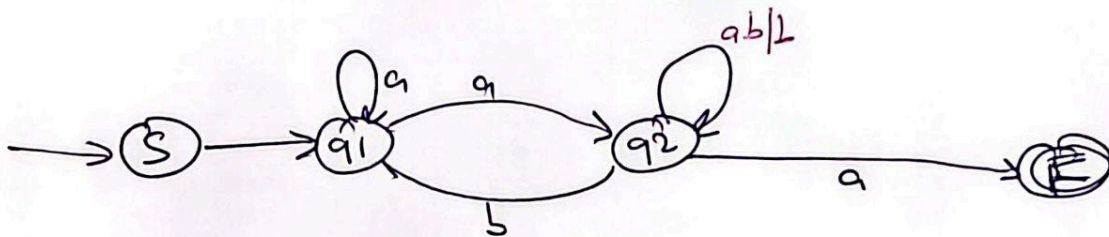
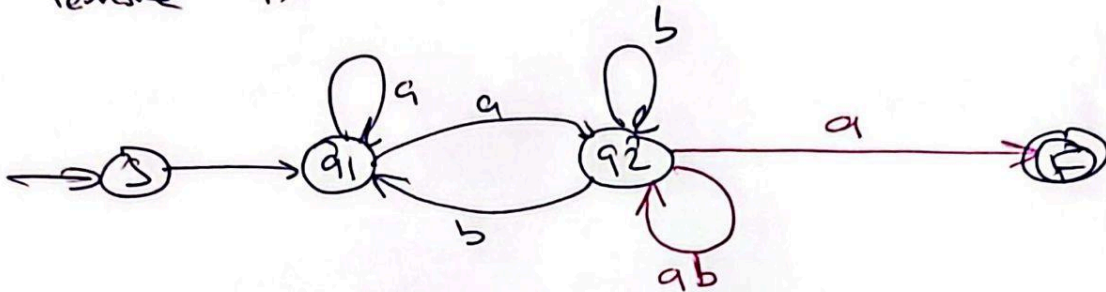


remove q_3

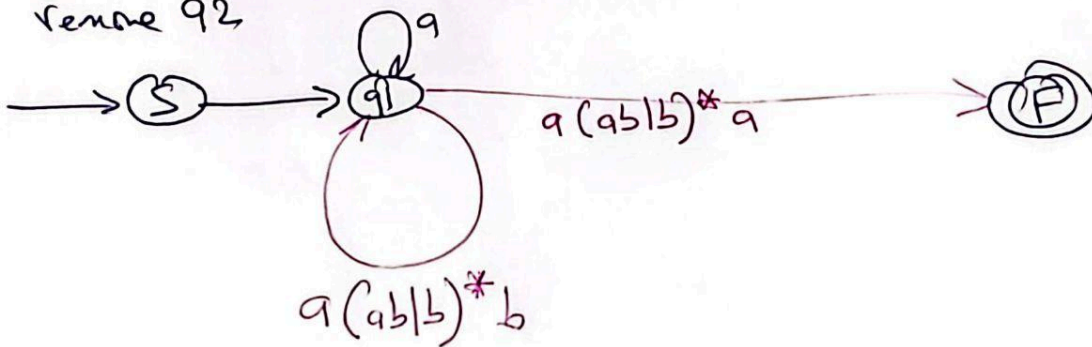




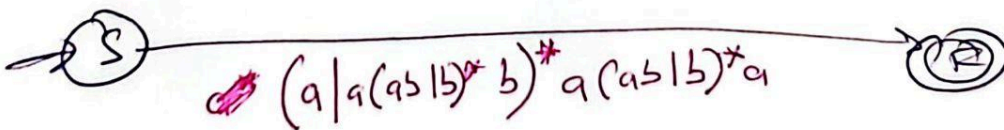
removing q3



removing q2



removing q1



Exercise 3 (4 points) [Regular Languages] :

Prove that the following languages are not regular:

1. [1 points] The number of 0 is not the same as the number of 1.

Use complement

Same Procedure as in Lecture/TD by choosing $s=0^p1^p$

2. [1 points] Let $\Sigma = \{0, 1, +, =\}$ and $ADD = \{x = y + z \mid x, y, z \text{ are binary integers, and } x \text{ is the sum of } y \text{ and } z\}$.

Assume it is regular, use pumping lemma for the string s is $1^{p+1}=10^p+1^p$

Since $|xy| \leq p$ and $|y| = k > 0$

By pumping y to generate more words,

We will have $1^{p+1+k}=10^p+1^p$ which is not correct. Therefore not in the language..

3. [2 points] $L = \{0^k \mid k \text{ is a prime number}\}$ is not regular.

We know there are infinite prime number

Let's choose the string $s=0^p$ such that p is a prime number and p is greater than the pumping length L .

Assume that $|y|=k$, Let's pump y , $p+1$ times, so that the word will be : $xy^{p+1}z$ The number of zeros will be $|xyz| + p|y| = p+pk$, therefore the generated word is 0^{p+pk}

But $p+pk = p(1+k)$ is not a prime number, therefore, the newly generated word is not in the language $\rightarrow$ Contradiction to the assumed hypothesis that if you pump, you will always get words in the language. Therefore, the language is not regular.

Exercise 4 (4 points) [CFG & PA] :

Generate the Context Free grammar for the following languages:

1. [2 point] The number of a is not the same as the number of b .

Means : a has more than b **OR** b has more than a

As in the TD :

$S \rightarrow A_{\text{more}} \mid B_{\text{more}}$

$A_{\text{more}} \rightarrow A_1 a A_1$

$A_1 \rightarrow b A_1 a \mid a A_1 b \mid A_1 A_1 \mid a A_1 \mid \epsilon$

$B_{\text{more}} \rightarrow \dots$ Same as A , but Swap A with B .

2. [2 point] The complement of the language $a^n b^n$ such that $n > 0$ (The big set of $\{a,b\}^*$)

We need to define the complement of $a^n b^n$:

Which is : $a^n b^m$ such that $n \neq m$ **Union** $\{a,b\}^* b a \{a,b\}^*$

$D \rightarrow S_{\text{previous_question}} \mid G b a G$

$G \rightarrow a G \mid b G \mid \epsilon$